Do Now · retrieval

Lesson 148 · Network Hardware: Switches, Routers and Access Points

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a firewall generally use to permit or block a connection?

Q2

What data structure follows FIFO?

Q3

What is the result of an AND gate with inputs 1 and 0?

Q4

What is a foreign key used for?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Local connections — A switch forwards Ethernet frames using learned MAC addresses.

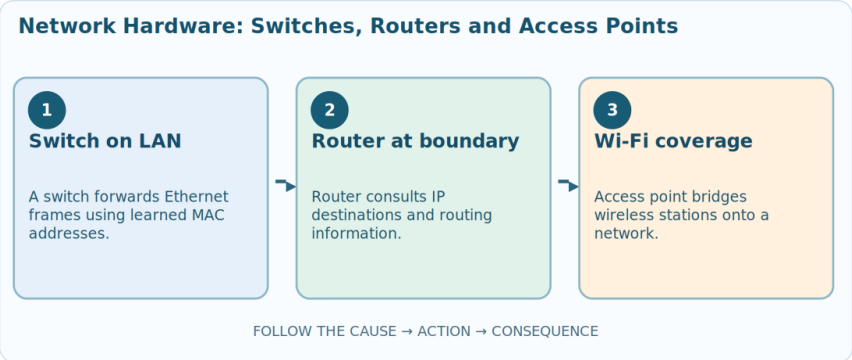
Core idea 2

Across networks — Router consults IP destinations and routing information.

Core idea 3

Wireless access — Access point bridges wireless stations onto a network.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.

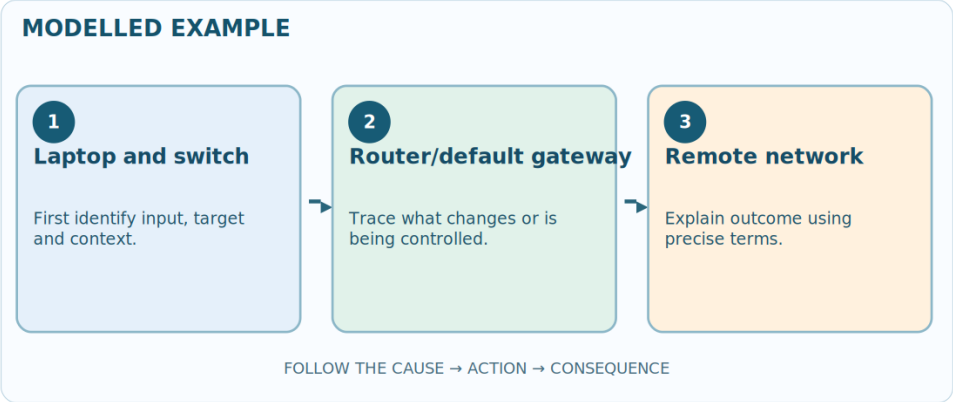
Follow the worked method

1. Laptop and switch — Identify the facts and make an initial prediction.

2. Router/default gateway — Explain the mechanism or trace the intermediate stage.

3. Remote network — Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Wired LAN works

Identify the facts and make an initial prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A basic Ethernet switch chooses internet routes using destination website names.

Use key vocabulary from memory

1. Explain switch in your own words.

2. Explain router in your own words.

3. Explain access point in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.